DEFORM DOE Optimization Loop: Lab Instructions

Use this handout with run_initial_doe.py on the lab computer. The workflow is staged on purpose: DEFORM runs are expensive, and each batch should be inspected before launching the next one.

Key Rule

Do not turn on every flag for a fresh study. Later stages need files created by earlier completed DEFORM runs. Run the workflow in passes.

What Stops the Loop?

The current code is not an unattended infinite loop. It runs one selected stage, then exits. The proposed-candidate batch is limited by NEXT_ROUND_NUMBER and MAX_ROUND_NUMBER.

```
RUN_NEXT_CANDIDATES = False
EXECUTE_NEXT_CANDIDATES = False
NEXT_ROUND_NUMBER = 2
MAX_ROUND_NUMBER = 2
```

Main Files

`run_initial_doe.py`	Spyder launcher for DOE generation, execution, processing, proposal, next-round execution, and reports.
`doe_runner.py`	Turns candidate rows into action CSVs/config JSONs and optionally runs DEFORM.
`process_doe_results.py`	Processes completed manifests into metrics and dataset.csv.
`dataset_builder.py`	Builds datasets from completed runs only. Planned/running/failed runs are ignored by default.
`visualize_doe_results.py`	Creates report/index.html, plots, and dataset_numeric_summary.csv.

Pass-by-Pass Workflow

1	Generate initial DOE inputs only. Inspect generated action/config files.	`PREPARE_RUNS=True EXECUTE_RUNS=False PROCESS_RESULTS=False PROPOSE_NEXT_CANDIDATES=False RUN_NEXT_CANDIDATES=False PROCESS_NEXT_CANDIDATES=False MAKE_REPORT=False`
2	Execute the initial DOE in DEFORM.	`PREPARE_RUNS=True EXECUTE_RUNS=True PROCESS_RESULTS=False PROPOSE_NEXT_CANDIDATES=False RUN_NEXT_CANDIDATES=False PROCESS_NEXT_CANDIDATES=False MAKE_REPORT=False`

3	Process completed initial runs, build dataset.csv, propose candidates_next.csv, and refresh report.	`PREPARE_RUNS=False EXECUTE_RUNS=False PROCESS_RESULTS=True PROPOSE_NEXT_CANDIDATES=True RUN_NEXT_CANDIDATES=False PROCESS_NEXT_CANDIDATES=False COMBINE_DATASETS=True MAKE_REPORT=True`

Next-Candidate Batch

4	Prepare the proposed candidate batch from candidates_next.csv. Inspect configs/actions.	`PREPARE_RUNS=False PROCESS_RESULTS=False PROPOSE_NEXT_CANDIDATES=False RUN_NEXT_CANDIDATES=True EXECUTE_NEXT_CANDIDATES=False PROCESS_NEXT_CANDIDATES=False MAKE_REPORT=False`
5	Execute the proposed next-candidate batch.	`PREPARE_RUNS=False PROCESS_RESULTS=False PROPOSE_NEXT_CANDIDATES=False RUN_NEXT_CANDIDATES=True EXECUTE_NEXT_CANDIDATES=True PROCESS_NEXT_CANDIDATES=False MAKE_REPORT=False`
6	Process round 02, update dataset_all.csv, propose another candidates_next.csv, and refresh reports.	`PREPARE_RUNS=False PROCESS_RESULTS=False RUN_NEXT_CANDIDATES=False PROCESS_NEXT_CANDIDATES=True COMBINE_DATASETS=True PROPOSE_NEXT_CANDIDATES=True MAKE_REPORT=True`

Expected Outputs

`candidates_initial.csv`	Pass 1	Initial Latin-hypercube DOE candidate table.
`manifest.csv`	Pass 1/2	Run tracking table. Only completed rows are used for dataset training.
`dataset.csv`	Pass 3 or 6	Per-study completed-run dataset with input features and metrics.
`dataset_all.csv`	Pass 3 or 6	Combined completed rows across initial DOE and next round. Bayesian training uses this when available.
`candidates_next.csv`	Pass 3 or 6	Bayesian proposal for the next batch.
`report/index.html`	MAKE_REPORT=True	HTML report with plots and summary table.

Important Dataset Behavior

dataset_builder.py filters manifest rows to status == completed by default. Planned, running, and failed runs are ignored so they do not train the surrogate or influence Bayesian proposal.

Troubleshooting

`No module named surrogate_model`	The lab code folder is missing surrogate_model.py or Spyder is not using the code folder as its working directory.
`Columns are missing: {...}`	Rebuild dataset.csv with the updated dataset_builder.py. The surrogate must train only on bounded input columns.

`No plots in report`	Install/enable matplotlib in the Spyder environment. The HTML summary still writes without plots.
`candidates_next.csv not found`	Run Pass 3 first with PROCESS_RESULTS=True and PROPOSE_NEXT_CANDIDATES=True.

Recommended Practice

After each pass, inspect the printed paths and the generated files before continuing. For new process variables or objective weights, keep N_CANDIDATES small until the output files and reports look correct.